MIL-R-40080C(ME)
21 May 1975
SUPERSEDING
MIL-R-40080B(ME)
29 November 1968

MILITARY SPECIFICATION

RAMP, LOADING, VEHICLE: PORTABLE, FOUR-SECTION, 12,000-POUND CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a free-standing 12,000-pound capacity, magnesium railcar-loading ramp which can be disassembled into four sections, stacked, and transported by forklift truck.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, and Seals.

Military

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

FSC 3990

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DRAWINGS

ME

TA13216E9207 - Ramp, Loading, Vehicle: Portable, Four-Section, 12,000-Pound Capacity.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The portable loading ramp (hereinafter called "ramp") shall be in accordance with Top Assembly TA13216E9207, and as specified herein. The ramp shall consist of four separate sections to facilitate assembly. Components used in the assembly shall include welded steel chain, shackle, hoist or sling hooks, and self-locking quick-release pins. Except as otherwise specified herein, the ramp shall be of magnesium all-welded construction with raised antiskid travel surface. The ramp shall terminate at railcar floor height.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the ramp. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more ramps as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce ramps that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Construction. Except as otherwise specified herein, the ramp shall be of all-welded construction. The ramp shall be constructed in accordance with the design, dimensions and materials shown on the applicable drawings and as specified herein. When tested as specified in 4.5.2.1 through 4.5.2.3, the ramp shall show no evidence of permanent deformation, damage or failure.

3.5 Assembly and disassembly. The ramp shall be capable of being assembled and disassembled as specified in 4.5.2.1 with the use of a standard military 4000-pound capacity forklift truck, or an equivalent forklift truck having the same general characteristics and approved by the contracting officer, using the forklift openings and forklift handles of Section 1 and the end openings of Sections 2, 3 and 4. The sections shall fit together, aline, and also readily disengage so that the quick-release pins can be fully inserted or withdrawn from the intended holes. The lifting-handle springs of Section 1 shall keep the lifting handles in the "up" position. The pin joints of the leg assemblies shall pivot freely. The use of tools shall not be required.

3.6 Stability. When securely attached to a railcar, the ramp shall be stable when tested as specified in 4.5.2.2.

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3.7 Capacity.

3.7.1 Rated load. The ramp shall have a rated load capacity of 12,000 pounds.

3.7.2 Overload. The ramp shall be capable of withstanding the stresses induced by loads of 200 percent of rated capacity without permanent deformation or indications of material or workmanship defects when tested as specified in 4.5.2.3.

3.8 Workmanship. Before and after painting, all parts and assemblies of the ramp shall be free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction ramp(s) shall be examined as specified in 4.5.1 Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. The preproduction ramp(s) shall be tested as specified in 4.5.2 through 4.5.2.3. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect. Inspection Level shall be II.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2 through 4.5.2.3. Inspection Level shall be S-3 with an AQL of 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The ramp shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Dimensions not as specified.
103. Components missing or not as specified.

Minor

201. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Assembly and disassembly. The ramp shall be assembled and disassembled on a firm, level surface with the use of a forklift truck as specified in 3.5. The ramp shall be evaluated for conformance with the requirements specified in 3.5. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.2 Stability. A forklift truck carrying a 4000-pound load as specified in 3.5, shall be operated up and down the ramp not less than 10 times after the ramp has been assembled to determine conformance with 3.6. Nonconformance to 3.6 shall constitute failure of this test.

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4.5.2.3 Overload. Each section of the assembled ramp under test shall be subjected to a static load of 24,000 pounds. This load shall be applied through an area extending not more than 9 inches on each side of the lateral centerline of the section. Each section shall remain under this load for 1 hour, after which the load shall be removed and the section shall be inspected to determine compliance with the requirements specified in 3.7.2. Nonconformance to 3.7.2 shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective and Inspection Level shall be S-2.

104.	Disassembly not as specified.
105.	Unpainted exterior metal surfaces not coated with preservative for Level A.
106.	Sections not placed and secured as specified for Level A or B.
107.	Strapping not as specified for Level A or B.
108.	Marking illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The ramp shall be disassembled into the four basic sections and the legs placed in the closed position and secured, as provided for. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss.

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5.1.1.2 Unprotected surfaces. Unpainted exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.2 Level C. The ramp shall be disassembled as specified in 5.1.1.1. Preservation and packaging shall be adequate to afford protection against deterioration and damage from the supplier to the initial destination. The supplier's standard commercial practice is acceptable when it fulfills these requirements.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. Each complete ramp, preserved and packaged as specified in 5.1, shall be prepared for shipment as a secured lift. The sections shall be placed on top of each other as follows: Section 1 shall be placed on the bottom followed by Sections 2 and 3 with Section 4 placed on top. The sections shall be secured together. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size 1-1/4 inch, 0.035 inch thick; and, unless otherwise specified herein, strapping shall be Finish B. When specified (see 6.2), Finish A strapping shall be used.

5.2.2 Level B. Each ramp shall be prepared for shipment as specified in 5.2.1, except strapping shall be Finish A.

5.2.3 Level C. Each ramp, preserved and packaged as specified in 5.1, shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ramp is intended for use on prepared or semi-prepared surfaces to facilitate forklift truck operation in the loading and unloading of railcars in open depot storage areas where permanent railcar docks are not provided. The ramp is sectionalized to facilitate assembly by a 4000-pound forklift. This ramp is capable of supporting a 4000-pound capacity forklift truck carrying a full capacity load.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of preproduction model, and number of ramps required (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(d) When strapping shall be Finish A (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production ramps from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish ramps conforming to this specification.

Custodian:

Army - ME

User activity:

Army - MT

Preparing activity:

Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-R-40080C(ME) Ramp, Loading, Vehicle: Portable, Four-Section, 12,000 Pound Capacity

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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